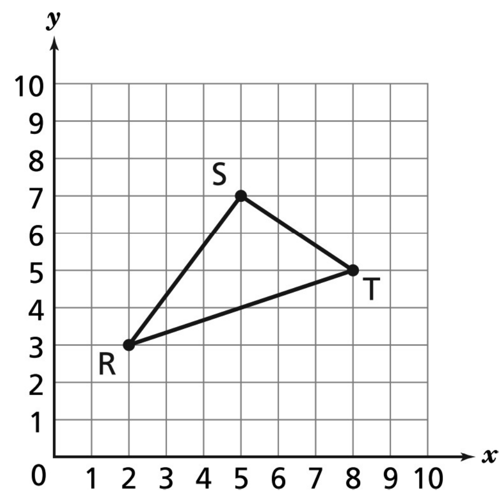
PAT 6 - coordinate system

Name: ______________________ **Date:** ______________

1. A triangle is plotted on the coordinate plane below.



Which coordinates represent, in order, the locations of point R, point S, and point T?

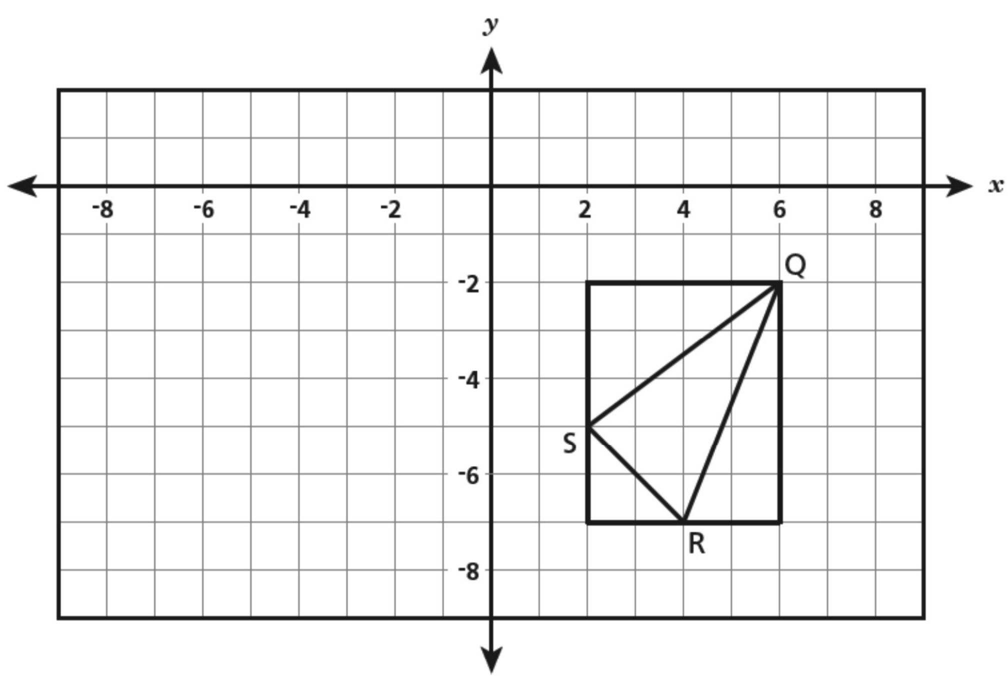
A. $(3, 2)$, $(7, 5)$, and $(5, 8)$

B. $(2, 3)$, $(7, 5)$, and $(8, 5)$

C. $(2, 3)$, $(5, 7)$, and $(8, 5)$

D. $(3, 2)$, $(5, 7)$, and $(5, 8)$

2. Triangle QRS, with vertices $Q(6, -2)$, $R(4, -7)$, and $5(2, -5)$, is drawn inside a rectangle, as shown below.



What is the area, in square units, of triangle QRS?

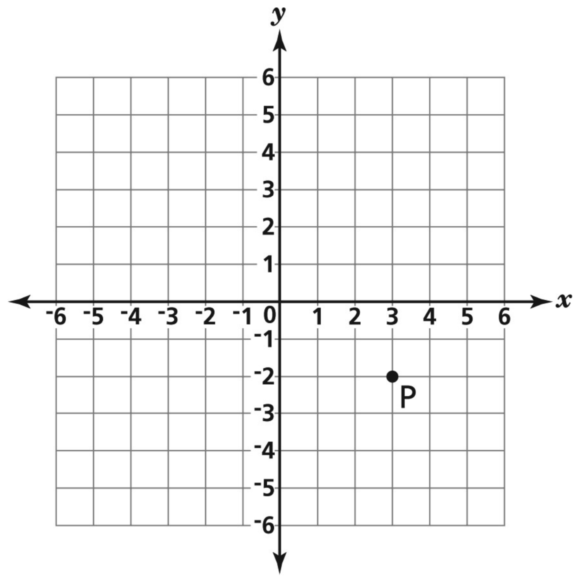
A. 7

B. 10

C. 13

D. 18

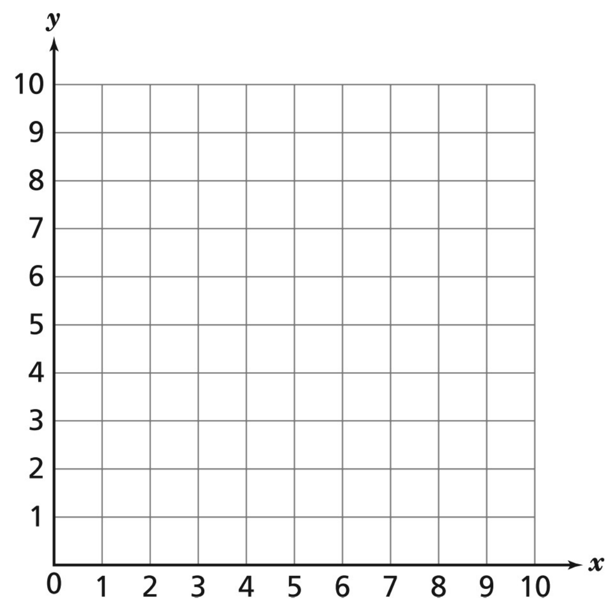
3. What are the coordinates of point P?



A. $(2, 3)$ B. $(3, 2)$ C. $(-2, 3)$ D. $(3, -2)$

4. On the grid below

- plot and label the points: $A(1, 5)$, $B(3, 2)$, $C(6, 2)$, $D(8, 5)$
- connect the points in order, starting with point A, to draw a quadrilateral



What type of quadrilateral is formed by connecting points A, B, C, and D?

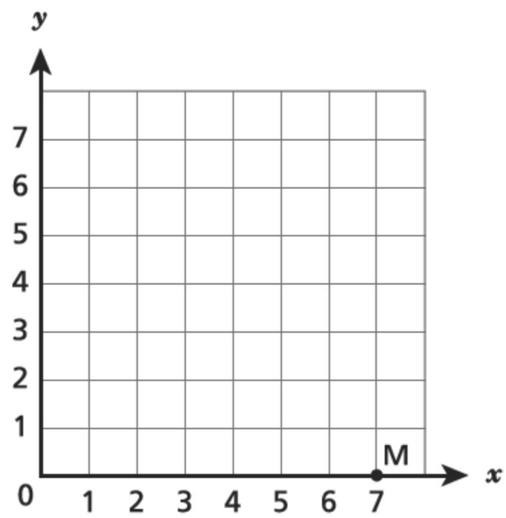
Explain how you determined the type of quadrilateral plotted on the grid.

5. A triangle has vertices on a coordinate grid at points $J(-1, 5)$, $K(4, 5)$, and $L(4, -2)$. What is the length, in units, of $\overline{KL}$?

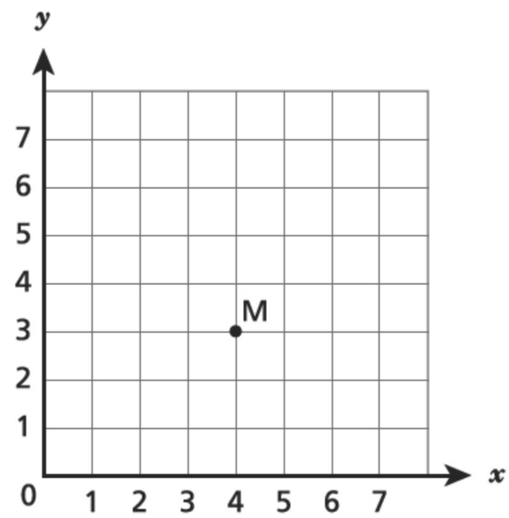
A. 3 B. 7 C. 8 D. 11

6. Which coordinate grid shows point M plotted at $(4, 3)$?

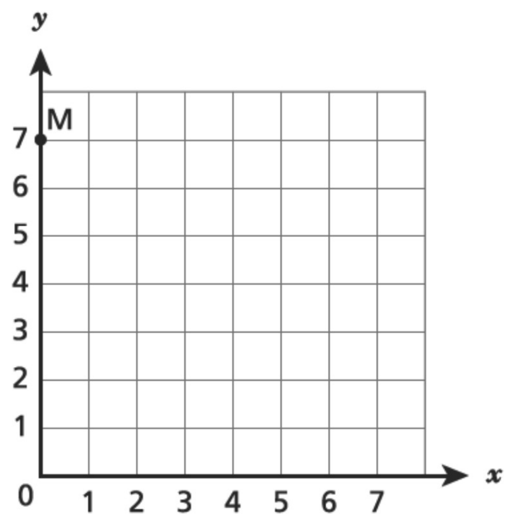
A.



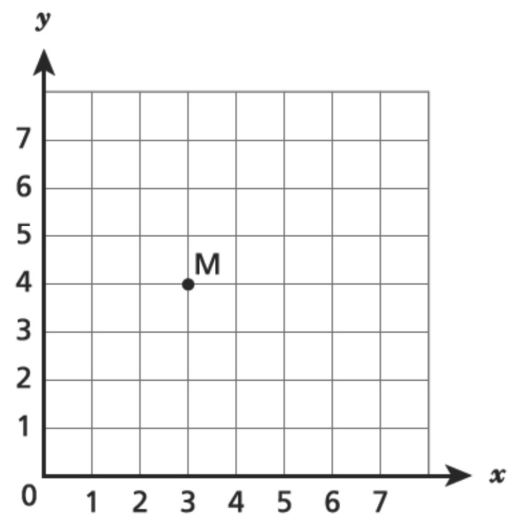
B.



C.



D.



7. Peter wants to plot the point $(2, 3)$ on a coordinate plane. Which statement describes how to plot this point starting from the origin?

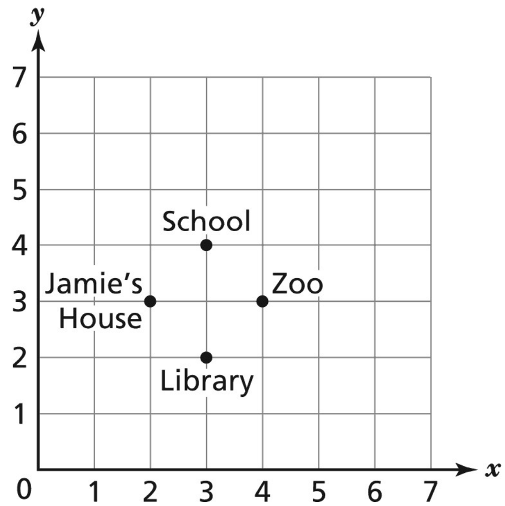
A. Move 2 units to the left and then 3 units down.

B. Move 3 units to the left and then 2 units down.

C. Move 2 units to the right and then 3 units up.

D. Move 3 units to the right and then 2 units up.

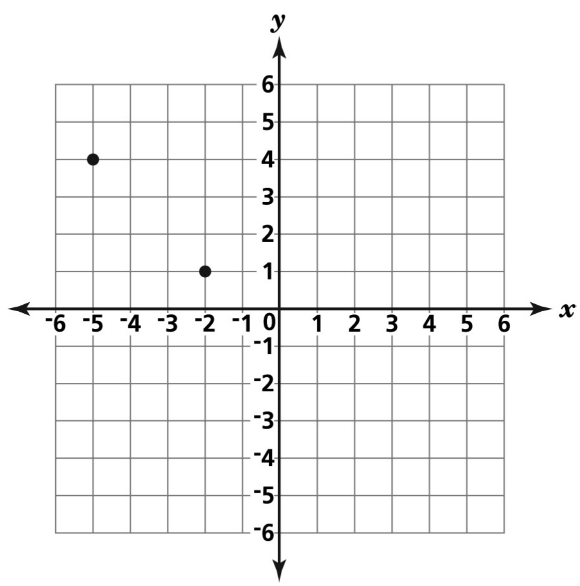
8. Jamie created a map for his friends. Each point on the map represents a different location.



What coordinates represent Jamie's house?

A. (2, 3) B. (3, 2) C. (3, 4) D. (4, 3)

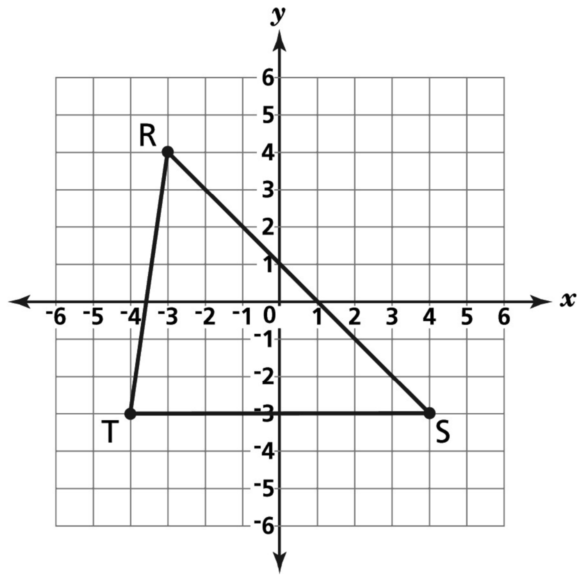
9. Two vertices of a right triangle are shown on the coordinate plane below.



Which point could represent the third vertex of the right triangle?

A. $(5, -1)$ B. $(1, -5)$ C. $(-1, 5)$ D. $(-5, 1)$

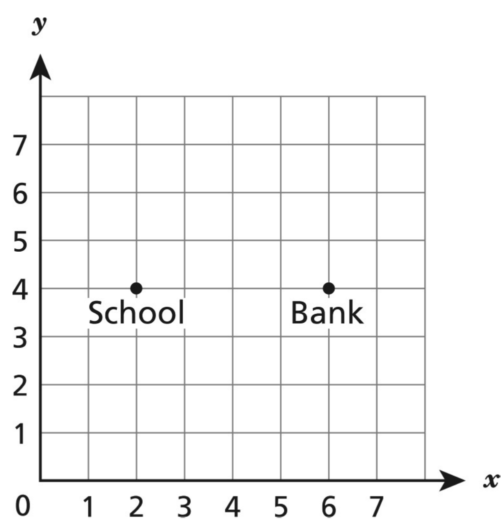
10. Triangle *RST* is shown on the coordinate grid below.



What are the coordinates of point *T*?

A. $(-3, -4)$ B. $(-3, 4)$ C. $(-4, -3)$ D. $(-4, 3)$

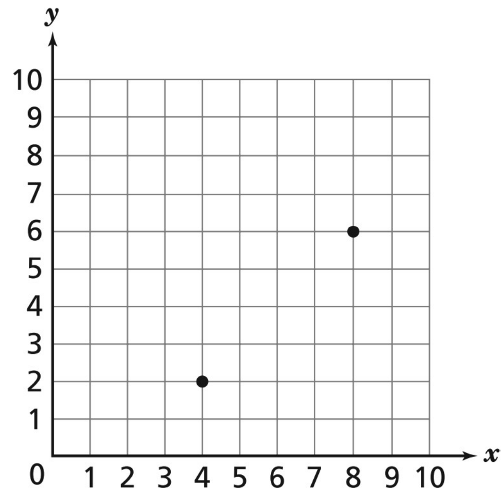
11. Mark graphed points on the coordinate plane below to represent the locations of his school and a bank.



Mark wants to add the location of the library on the coordinate plane. The distance from the library to the school is the same as the distance from the bank to the school. Which ordered pair could be the coordinates of the library?

A. $(2, 4)$ B. $(2, 8)$ C. $(4, 4)$ D. $(6, 8)$

12. Carlos plots two points on the grid below.



He wants to plot two more points and then connect all four points to form a square. Which two points should Carlos plot to form a square?

A. $(4, 2)$ and $(8, 6)$ B. $(4, 6)$ and $(6, 6)$ C. $(4, 2)$ and $(6, 2)$ D. $(4, 6)$ and $(8, 2)$

13. The coordinates of the points below represent the vertices of a rectangle.

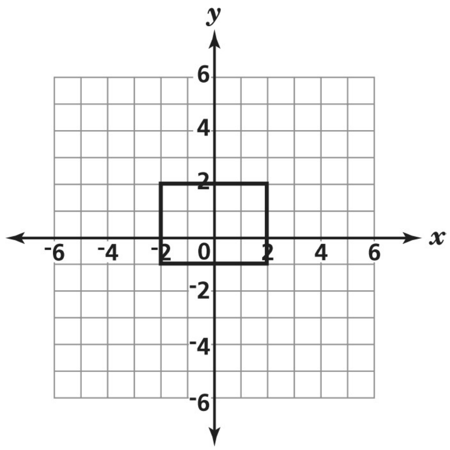
P: $(2, 2)$
Q: $(6, 2)$
R: $(6, 5)$
S: $(2, 5)$

What is the perimeter, in units, of rectangle *PQRS*?

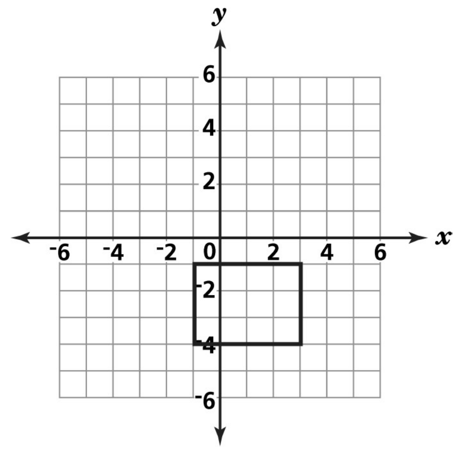
A. 8 B. 12 C. 14 D. 16

14. Which figure below represents a rectangle with vertices $(3, 2)$, $(-1, 2)$, $(-1, -1)$, and $(3, -1)$?

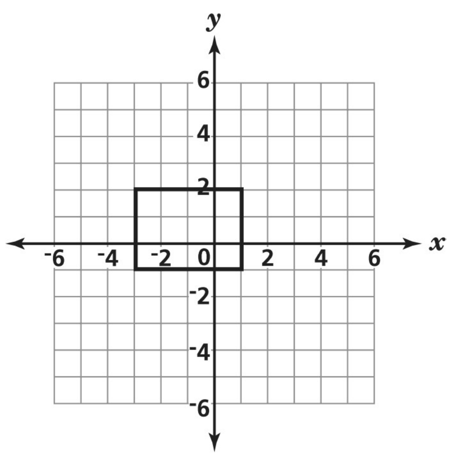
A.



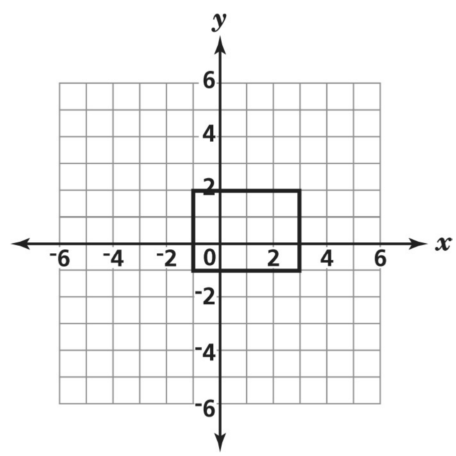
B.



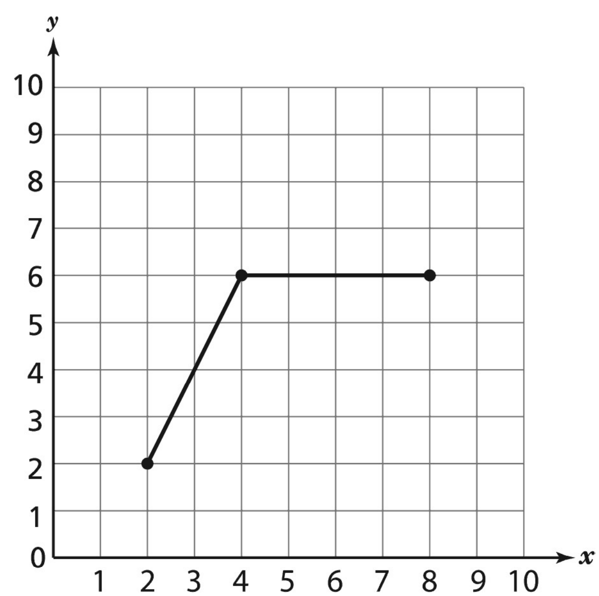
C.



D.



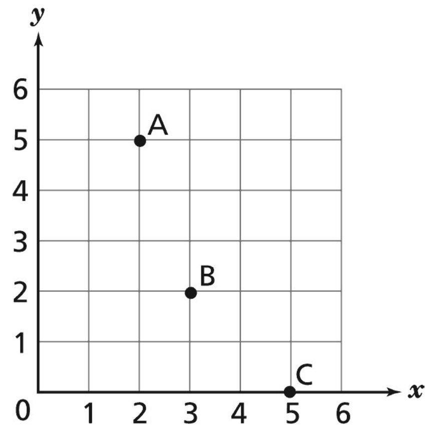
15. Michael plots three points on the grid below and connects the points.



What coordinates should Michael plot next if he wants to draw a parallelogram?

A. $(10, 2)$ B. $(6, 2)$ C. $(2, 8)$ D. $(2, 6)$

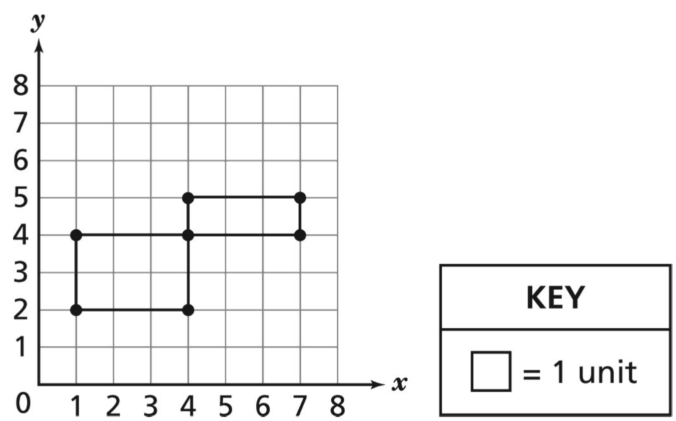
16. Ellen plotted three points on the grid below.



What are the coordinates of point C?

Explain how you determined the coordinates of point C.

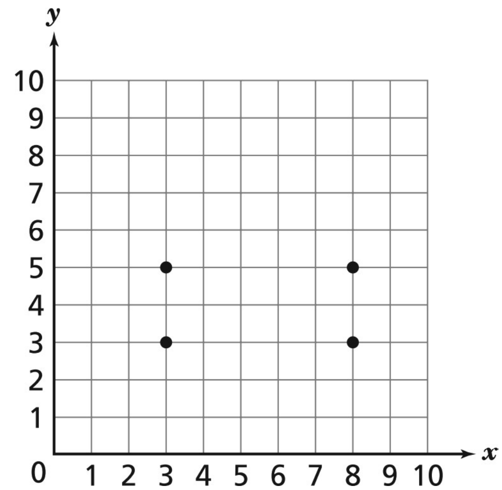
17. Mitchell drew two rectangles on the grid below.



What is the total perimeter, in units, of the two rectangles?

A. 8 B. 9 C. 10 D. 18

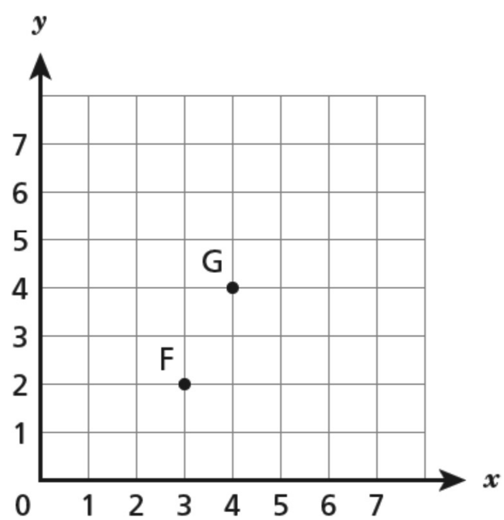
18. Ned wants to draw a pentagon on the grid below by plotting a fifth point and then connecting all of the points.



Which coordinates would *not* complete the pentagon?

A. (5, 8) B. (6, 7) C. (7, 2) D. (8, 4)

19. Points F and G have been plotted on the coordinate plane below.



Point G and point H are the same distance from point F. Which coordinates could be the location of point H?

A. (1, 2) B. (4, 2) C. (5, 1) D. (2, 5)